Coursework Description Sheet

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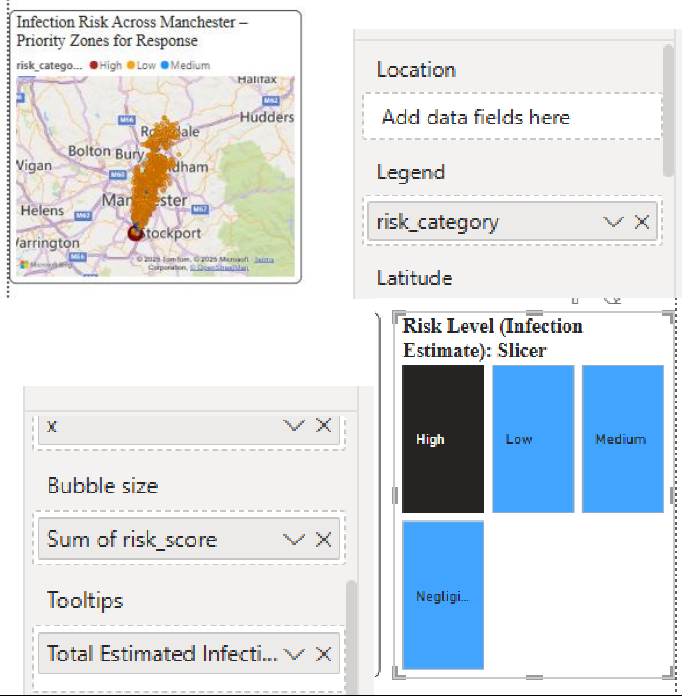
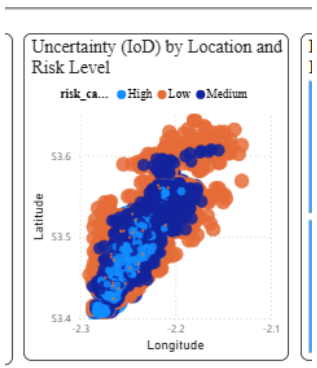
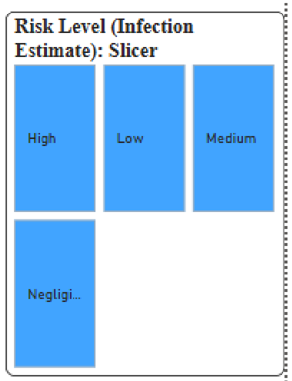
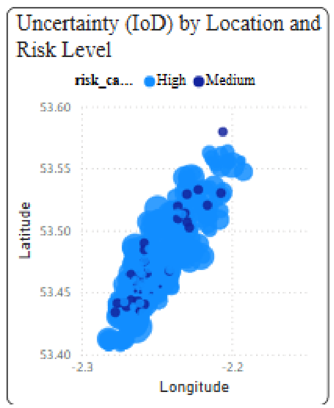
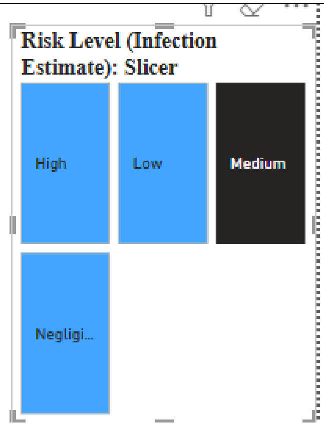
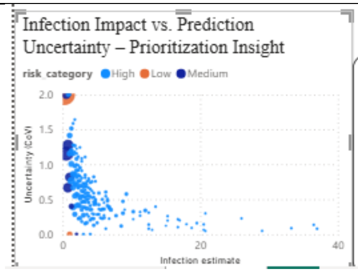
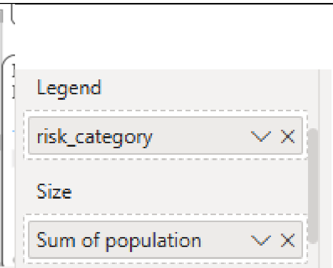
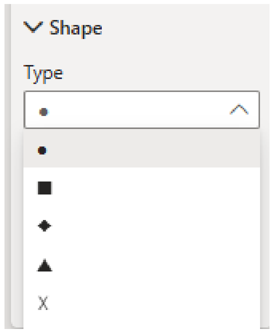
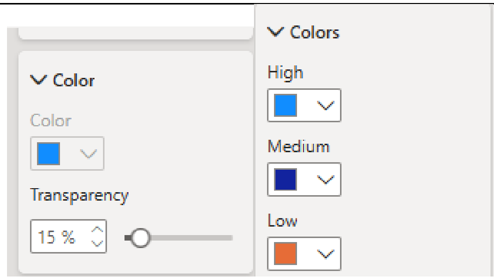
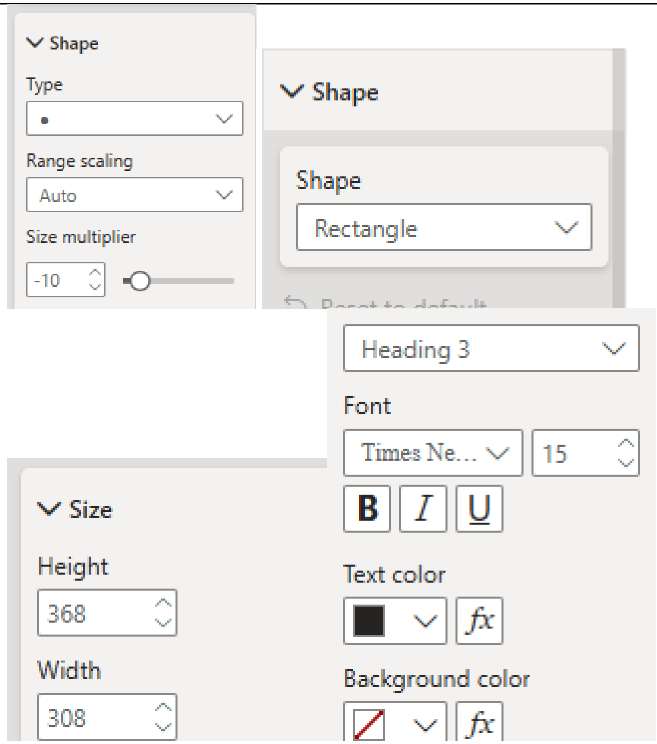
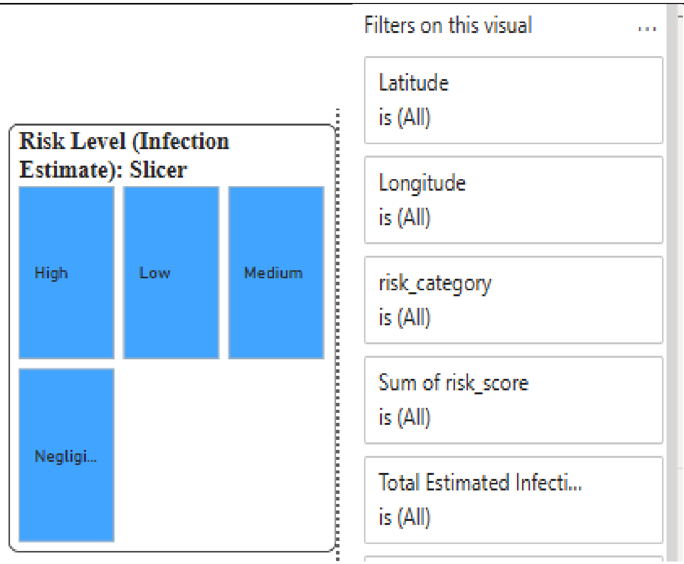
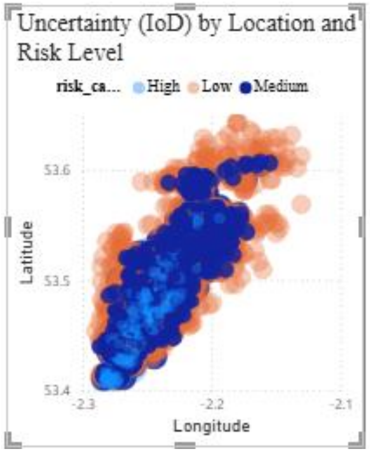
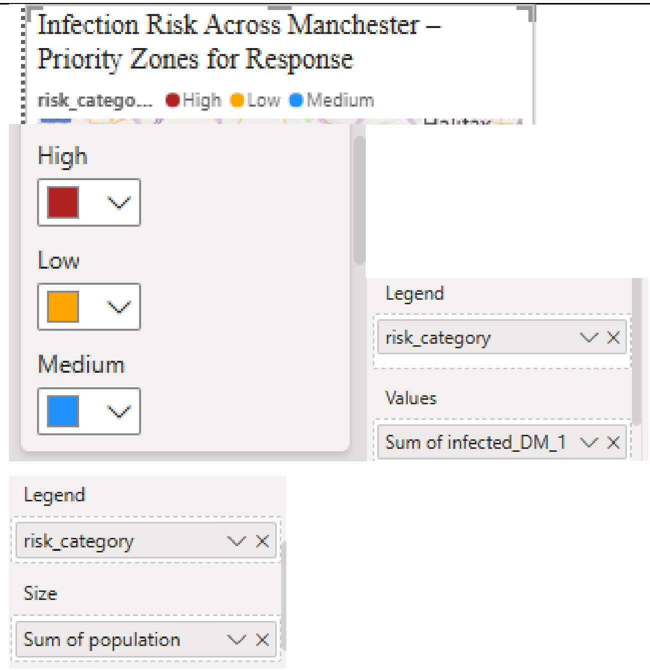
Question	Description	Figure
Fit to Task/User needs		
Location task - How does the visualisation allow users to access the spread of airborne infectious disease outbreak as per their risk score across Manchester?	• A filled map is developed using Power BI, and the map helps in identifying the high-risk areas. • The bubble size in the map is dependent on the risk_score variable. • The Risk level (infection estimate) slicer is used here to identify the high-risk infection area on the map. • The user can determine the infected areas on the map based on the u category. • Tooltips are used here to display textual elements like values and titles after hovering the mouse cursor.	

		
Time task - How does the visualization allow user to understand the spread of airborne diseases over time?	• In the dataset, there is no column to contain the date for exploring the infectious disease. • Although it is not possible to develop a time-series plot, the scatter plot helps users evaluate temporal spread patterns. • High infection variability at a location indicates a rapid change in infection rates. • The plot can easily determine high uncertainty that allows the user to predict rapidly changing or currently active zones. • The scatter plot can easily spot the new locations with increasing uncertainty. • The slicer allows for detecting the spatial-temporal infection based on the Risk level (infection estimate).	

		 
Multi-dimensional data task - How does the visualization allow user to identify correlation amongst the impact and uncertainty?	• The x-axis of the scatter plot represents the estimated infection impact per location. • The y-axis is used here to show the coefficient of variation for analysing the relative uncertainty of infection. • The colour of the scatter plot is dependent on the risk category, and each point of the scatter plot represents the population segment. • The colour coding in the plot is used to differentiate the high-risk and low-risk areas based on the collected data. • In the scatter plot, a negative correlation has been observed, and higher impact areas tend to have lower uncertainty.	  
Use of colour - How does the use of	• Relevant colour schemes have been applied to	

colour in this dashboard enhance the readability and effectiveness of the data presentation?	denote the risk category. ● All the colours are selected for mitigating the colourblind issue. ● Each colour has readability and is distinct from the other colours. ● Distinct colours assist in differentiating the risk categories based on the infectious disease.	
Use of graphic design principles - How does the application of graphic design principles enhance the clarity and effectiveness of the data presentation in this dashboard?	● In the dashboard, appropriate font size, font style and colours are used to improve the consistent design. ● Consistent font style and font size have been used here to create a sense of similarity to the user. ● A map visual allows for making decisions regarding infectious disease, and the user can get data by hovering over the visual. ● In the slicer, the rectangle shape is selected, and in the scatter plot, "." is used in a distinct colour for understanding the impact, risk, and uncertainty of infectious disease.	

Use of interaction - How does the use of interactive design elements improve the user's ability to explore and interpret data on this dashboard?	• The filter option is used here to improve the simplicity of the data visualisation. • A slicer has been applied to explore the infected areas based on the categorical variables. • Detailed tooltips are applied to provide information about the visuals.	 

Use of text and legend - How do the use of text and legends contribute to the clarity and user comprehension of the data presented in this dashboard?	● Legends are shown on the top of the visuals, and it help the user interpret data. ● Slicer is used here to easily evaluate the infectious disease based on the u category. ● In the map visual, risk_score is used as the legend to explore the high, medium, and low risk category areas. ● The plot title has been added in each visual, and the title helped in improving the data presentation in the dashboard.	

References

- Nielsen, P.V. and Xu, C. (2021). Multiple airflow patterns in human microenvironment and the influence on short-distance airborne cross-infection – A review. *Indoor and Built Environment*, 31(5), pp.1161–1175. https://doi.org/10.1177/1420326x211048539.
- Schöttler, S., Yang, Y., Pfister, H. and Bach, B. (2021). Visualizing and Interacting with Geospatial Networks: A Survey and Design Space. *Computer Graphics Forum*. https://doi.org/10.1111/cgf.14198.